

WEYBURN, SK, Canada

WMO: 714520

Lat: 49.700N

Lon: 103.800W

Elev: 1931

StdP: 13.70

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-20.7	-31.8	1.0	-25.6	-26.3	1.4	-20.6	35.0	7.7	30.9	10.3	8.2	310	0.734

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	89.0	66.4	85.3	65.5	81.8	64.0	71.8	82.1	69.0	79.3	66.6	77.4	12.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	112.4	77.1	65.4	100.9	74.2	63.0	92.4	71.2	36.9	82.6	34.4	78.8	32.3	77.4	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.0	25.5	22.8	-32.7	96.5	6.1	4.0	-37.2	99.4	-40.7	101.7	-44.2	104.0	-48.7	106.9
			-32.7	75.3	6.3	3.1	-37.3	77.5	-41.0	79.3	-44.5	81.1	-49.1	83.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	7.3	10.1	23.0	39.6	51.8	61.2	67.1	65.5	55.6	41.0	24.9	11.8
	DBStd	24.01	15.31	14.76	14.43	10.63	8.19	6.43	5.65	6.37	8.36	9.40	12.46	14.44
	HDD50	5973	1324	1117	839	333	78	3	0	0	38	304	754	1183
	HDD65	9963	1789	1537	1303	763	415	147	42	72	298	745	1204	1648
	CDD50	1734	0	0	1	20	133	340	529	480	206	24	1	0
	CDD65	248	0	0	0	0	5	33	107	87	16	0	0	0
	CDH74	3469	0	0	0	14	155	468	1249	1216	352	15	0	0
	CDH80	1140	0	0	0	1	29	118	426	457	107	2	0	0
Wind	WSAvg	10.8	11.3	10.9	11.8	12.1	12.0	10.5	9.4	9.4	10.3	10.9	10.7	10.7
Precipitation	PrecAvg	15.4	0.6	0.5	0.6	1.2	2.2	2.6	2.2	1.9	1.7	0.8	0.4	0.6
	PrecMax	22.8	1.7	1.2	2.0	4.3	6.7	9.2	7.6	7.8	4.3	2.6	0.9	1.3
	PrecMin	9.9	0.1	0.1	0.0	0.1	0.0	0.6	0.2	0.0	0.2	0.0	0.0	0.2
	PrecStd	3.5	0.3	0.3	0.5	0.9	1.5	1.8	1.7	1.6	1.1	0.7	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	42.8	60.5	75.5	83.8	88.8	93.8	94.9	88.3	75.8	60.0	44.1
		MCWB	37.3	38.0	46.7	55.0	58.9	66.5	69.4	68.4	62.6	56.2	47.8	38.3
	2%	DB	37.2	37.6	51.8	67.9	78.3	83.3	88.4	89.7	83.2	67.7	50.8	39.0
		MCWB	34.1	35.1	42.9	51.2	57.5	65.0	68.6	65.4	61.2	52.8	42.6	35.4
	5%	DB	33.6	33.8	44.9	63.0	73.7	79.2	84.6	85.1	78.0	61.3	45.1	35.0
		MCWB	31.7	32.3	39.3	48.6	55.1	62.4	67.6	65.0	59.2	49.5	39.1	32.8
	10%	DB	28.7	30.6	39.5	57.5	68.9	75.6	81.1	81.0	72.2	56.0	40.3	31.0
		MCWB	27.4	29.3	36.3	45.9	52.9	61.0	66.2	63.3	57.0	46.9	36.4	29.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	38.3	48.6	56.6	63.0	73.8	75.7	73.0	68.0	57.9	48.7	38.7
		MCDB	41.1	41.7	58.2	70.9	76.0	84.1	84.7	86.4	78.0	71.5	60.4	42.5
	2%	WB	34.3	35.0	43.8	52.8	59.6	68.4	72.8	69.7	63.9	54.1	43.4	35.3
		MCDB	36.8	37.4	51.4	65.4	71.7	76.4	82.8	81.3	77.6	65.2	50.0	37.9
	5%	WB	31.8	32.4	39.5	49.7	57.1	65.1	70.2	67.3	61.0	50.6	39.6	32.6
		MCDB	33.7	33.7	44.1	60.4	69.1	74.7	79.7	79.9	74.1	59.4	44.1	34.8
	10%	WB	27.7	29.3	36.3	46.7	54.9	62.9	67.8	65.2	58.2	47.7	36.6	29.1
		MCDB	29.0	30.6	39.4	56.1	66.2	72.1	78.0	77.0	69.9	55.1	40.0	30.7
Mean Daily Temperature Range	5% DB	MDBR	18.1	18.2	17.5	22.0	24.3	21.7	23.9	25.7	24.9	20.9	17.8	17.0
		MCDBR	21.1	19.0	24.1	32.2	32.5	28.6	29.6	32.7	34.6	31.5	25.1	20.7
		MCWBR	18.6	16.5	15.9	17.6	15.4	13.2	13.3	13.2	15.7	18.0	17.1	17.1
	5% WB	MCDBR	20.9	18.5	22.2	30.1	27.2	24.2	25.5	27.4	29.6	28.4	23.7	19.7
		MCWBR	18.7	16.0	15.2	17.4	14.2	13.2	13.8	13.1	14.8	16.9	16.6	16.8
Clear-Sky Solar Irradiance	taub	0.247	0.270	0.272	0.327	0.340	0.352	0.361	0.373	0.325	0.285	0.254	0.244	
	taud	2.370	2.339	2.507	2.398	2.413	2.422	2.416	2.358	2.464	2.551	2.503	2.403	
	Ebn at Noon	262	281	302	292	291	287	283	273	278	275	260	247	
	Edh at Noon	20	27	27	34	35	35	35	35	28	22	17	17	
All-Sky Solar Radiation	RadAvg	441	785	1201	1565	1851	1940	2107	1680	1258	756	434	319	
	RadStd	42	70	95	106	138	127	115	121	114	86	60	31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	-2.11	-2.48	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page